Inverse Functions & More Logarithmic Derivatives

MATH 145 – Applied Calculus

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Derivatives of the form $f(x)^{g(x)}$

We know how to find $\frac{d}{dx}x^a$ when a is a constant:

$$\frac{d}{dx}x^a = ax^{a-1}$$

We also know now how to find $\frac{d}{dx}a^x$ when a is a constant:

$$\frac{d}{dx}a^x = a^x \ln(a)$$

Derivatives of the form $f(x)^{g(x)}$

What is the derivative of $\dfrac{d}{dx}x^x$?

$$\begin{aligned}
\frac{d}{dx}x^x &= \frac{d}{dx}e^{\ln(x^x)}\\
&= e^{x\ln(x)}\\
&= e^{x\ln(x)}\left(\frac{d}{dx}\left(x\ln\left(x\right)\right)\right)\\
&= e^{\ln(x^x)}\left(x\left(\frac{d}{dx}\left(\ln\left(x\right)\right)\right)+\ln\left(x\right)\left(\frac{d}{dx}x\right)\right)\\
&= x^x\left(x\left(\frac{1}{x}\right)+\left(\ln\left(x\right)\right)\left(1\right)\right)\\
&= x^x(\ln\left(x\right)+1)
\end{aligned}$$

Example: Derivatives of the form $f(x)^{g(x)}$

1. Find $\dfrac{d}{dx}(x^2+1)^{(x^2+1)}$.

2. Find $\dfrac{d}{dx}\sin(x)^{\sin(x)}$.

Derivatives of the form $f(x)^{g(x)}$

What is the derivative of $\dfrac{d}{dx}f(x)^{g(x)}$?

$$\begin{aligned}
\frac{d}{dx}f(x)^{g(x)} &= \frac{d}{dx}e^{\ln\left(f(x)^{g(x)}\right)} \\
&= e^{g(x)\ln(f(x))} \\
&= e^{g(x)\ln(f(x))}\left(\frac{d}{dx}\left(g(x)\ln\left(f(x)\right)\right)\right) \\
&= e^{\ln\left(f(x)^{g(x)}\right)}\left(g(x)\left(\frac{d}{dx}\left(\ln\left(f(x)\right)\right)\right)+\ln\left(f(x)\right)\left(\frac{d}{dx}g(x)\right)\right) \\
&= f(x)^{g(x)}\left(g(x)\left(\frac{f'(x)}{f(x)}\right)+\left(\ln\left(f(x)\right)\right)g'(x)\right) \\
&= f(x)^{g(x)}\left(g(x)\left(\frac{f'(x)}{f(x)}\right)+g'(x)\ln\left(f(x)\right)\right)
\end{aligned}$$

Example: Derivatives of the form $f(x)^{g(x)}$

1. Find $\dfrac{d}{dx}(x^2+1)^{x^3}$.

2. Find $\dfrac{d}{dx}\sin(x)^{\cos(x)}$.

Inverse Functions

Inverse Functions

Given a function $f : A \to B$, the **inverse function** of $f(x)$ is the function $f^{-1} : B \to A$ such that

$$f(f^{-1}(x)) = x \qquad \text{and} \qquad f^{-1}(f(x)) = x$$

Inverse Functions

Only **one-to-one** functions, functions for which if $f(x) = f(y)$, then $x = y$, for all x and y in the domain.

One-to-one functions must pass the **horizontal line test**.

If a function is not one-to-one, you can sometimes restrict the domain to get a one-to-one function.

The graph of the inverse function f^{-1} is the graph of f reflected over the line $y = x$.

Inverse Trigonometric Functions

By restricting the domain, the trigonometric can be made to have inverse function.

For instance, if we restrict the domain of the sine function to $[-\frac{pi}{2}, \frac{pi}{2}]$, then we can define the inverse function $\sin^{-1} : [-1, 1] \to [-\frac{pi}{2}, \frac{pi}{2}]$, also known as $\arcsin(x)$.

Similarly, if we restrict the domain of the tangent function to $[-\frac{pi}{2}, \frac{pi}{2}]$, the we can define the inverse function $\tan^{-1} : [-\infty, \infty] \to [-\frac{pi}{2}, \frac{pi}{2}]$, also known as $\arctan(x)$.

The derivatives of these inverse functions are very important in the study of *antiderivatives* because $\sin^2(x) + \cos^2(x) = 1$.

Inverse Trigonometric Functions

To find the derivative of $\sin^{-1}$, we will use a trick built around the fact that $\sin(\sin^{-1}(x)) = x$, and the fact that $\sin^2(x) + \cos^2(x) = 1$ implies $\cos(x) = \sqrt{1 - \sin^2(x)}$.

$$\begin{aligned}
\frac{d}{dx} x &= 1 \\
\frac{d}{dx} \sin(\sin^{-1}(x)) &= 1 \\
\cos(\sin^{-1}(x)) \left(\frac{d}{dx} \sin^{-1}(x) \right) &= 1 \\
\frac{d}{dx} \sin^{-1}(x) &= \frac{1}{\cos(\sin^{-1}(x))} \\
\frac{d}{dx} \sin^{-1}(x) &= \frac{1}{\sqrt{1 - (\sin(\sin^{-1}(x)))^2}} \\
\frac{d}{dx} \sin^{-1}(x) &= \frac{1}{\sqrt{1 - x^2}}
\end{aligned}$$

Example: Inverse Trigonometric Functions

1. Find $\dfrac{d}{dx}\cos^{-1}(x)$.

2. Find $\dfrac{d}{dx}\tan^{-1}(x)$.

3. Find $\dfrac{d}{dx}\sqrt{x}\tan^{-1}(x^2)$.

4. Find $\dfrac{d}{dx}e^{\sin^{-1}(\ln(x))}$.

Derivatives of Inverse Trigonometric Functions

- $\dfrac{d}{dx}\sin^{-1}(x) = \dfrac{1}{\sqrt{1-x^2}}$.

- $\dfrac{d}{dx}\cos^{-1}(x) = -\dfrac{1}{\sqrt{1-x^2}}$.

- $\dfrac{d}{dx}\tan^{-1}(x) = \dfrac{1}{1+x^2}$.